



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING
工程學院
機械工程學系

Reference Letter of Zhao Xuecen Jack

To Whom It May Concern,

I am pleased to recommend Mr. ZHAO Xuecen Jack, a Bachelor of Engineering student in Mechanical Engineering at The Hong Kong Polytechnic University. I served as his supervisor for his inter-departmental Final Year Project entitled “*Design and Development of an Electric Toy Car Powered by a Fuel Cell Stack.*”

Jack played a key role in the design and integration of *Vandi*, an autonomous toy vehicle incorporating a passive vanadium e-fuel cell stack, a hybrid power system, ROS-based control, LiDAR, camera vision, AI-assisted perception, and a browser-based frontend interface. The project was highly ambitious in scope and required the team to deliver a fully functional interdisciplinary system within a limited timeframe. Despite these challenges, the team achieved a successful and impressive outcome.

From a mechanical engineering perspective, Jack was responsible for the design of the fuel cell stack geometry, vehicle packaging, manufacturing coordination, and overall system integration. He also carried out SolidWorks modelling and simulations to support the design process. He demonstrated a strong awareness of Design for Manufacturing (DfM) and Design for Assembly (DfA), particularly in addressing issues related to component fastening, sealing, assembly sequencing, and integration with the vehicle structure.

Beyond his core mechanical responsibilities, Jack actively contributed to areas outside his primary discipline. He was involved in ROS-based control development, AI and vision implementation, frontend design, and interdisciplinary collaboration across mechanical, electrical, and biomedical engineering domains. His ability to bridge mechanical design with software, electronics, energy systems, and AI significantly strengthened the overall project.

The Vandi project presented numerous real-world engineering challenges, including packaging constraints, fuel-cell leakage risks, corrosion, limited passive power density, electrical integration, and system testing. Jack approached these challenges with persistence and sound engineering judgement. He demonstrated



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the ability to identify problems, coordinate effectively with teammates, refine design solutions, and contribute to delivering a coherent and working prototype.

Overall, Jack has demonstrated creativity, technical independence, and strong problem-solving skills. His performance reflects a solid engineering mindset and an ability to quickly learn and integrate knowledge across disciplines. I am confident in his potential and am pleased to recommend him for further academic, research, or professional opportunities.

Yours sincerely,

Maochun WU, Ph.D.

Assistant Professor

Department of Mechanical Engineering

The Hong Kong Polytechnic University